Probabilistic Combinatorics

Sheet 6

May 27, 2024

Problem 1. For a graph $G = (V, E)$ let $\omega(G)$ be the biggest number ℓ, such that G contains a full subgraph K_ℓ. Furthermore let $X_\ell(G)$ be the random variable on $G(n, 1/2)$, which assigns to a graph the number of complete subgraphs K_ℓ of G. Proof the following statements

(i)

$$\mathbb{E}(X_\ell) = \binom{n}{\ell} 2^{-\binom{\ell}{2}}$$

(ii) For every $\varepsilon > 0$:

$$\lim_{n\to\infty} \mathbb{P}(\omega \geq (2+\varepsilon)\log n) = 0$$

Proof.
It is known that $G(n, 1/2)$ is the discrete probability space where the sample space Ω_n is the set of all undirected graphs $G = ([n], E)$ while the probability space is $p_n(G) = \frac{1}{2}^{|E|} \frac{1}{2}^{n-|E|}$ for any graph $G \in \Omega_n$.

Let S be the set of all subsets of vertices in G with cardinality ℓ. For any $s \in S$, let G_s be the induced subgraph of a graph $G \in \Omega_n$ whose vertex set is s. Next, we define the indicator random variable $X_\ell^{(s)} : \Omega_n \to \mathbb{N}$ defined by the following rule for any graph $G \in \Omega_n$

$$X_\ell^{(s)} = \begin{cases} 1 \text{ if } G_s \text{ is } K_\ell \\ 0 \text{ otherwise} \end{cases}$$

That is, it is obvious that $X_\ell = \sum_{s\in S} X_\ell^{(s)}$. Now, we define the indicator variable $X_\ell = 1_{A_\ell^{(s)}}$ of the event

$$A_\ell^{(s)} = \{G \in \Omega_n | G_s \text{ is } K_\ell\}$$

then $\mathbb{E}(X_\ell^{(s)}) = \mathbb{E}(1_{A_\ell^{(s)}}) = \mathbb{P}(A_\ell^{(s)})$. Recall that $X_\ell = \sum_{s\in S}(X_\ell^{(s)})$, by linearity of expectation, we have

$$\mathbb{E}(X_\ell) = \sum_{s\in S} \mathbb{E}(X_\ell^{(s)}) = \sum_{s\in S} \mathbb{P}(A_\ell^{(s)}) = |S|\mathbb{P}(A_\ell^{(s)}) \tag{1.1}$$

That is, it remains to find $\mathbb{P}(A_\ell^{(s)})$ and $|S|$ to calculate $\mathbb{E}(X_\ell)$. It is known that, for a set $s \in S$, which is a subset of vertices in G with cardinality ℓ, and a graph $G \in \Omega_n$, the graph G_s is K_ℓ when there exists an edge between any pair of distinct vertices from ℓ vertices in s. That is, such $\binom{\ell}{2}$ edges must be presented in G. Moreover, it is known that the probability to choose a graph $G \in \Omega_n$ having a specific edge is $\frac{1}{2}$. Therefore, $\mathbb{P}(A_\ell^{(s)}) = 2^{-\binom{\ell}{2}}$. Moreover, it is obvious that $|S| = \binom{n}{\ell}$. It follows from the equation (1.1) that

$$\mathbb{E}(X_\ell) = \binom{n}{\ell} 2^{-\binom{\ell}{2}}$$

and the proof is finished for the statement (i).

In order to prove the statement (ii), we want to show the following equality holds for $\ell \geq (2+\varepsilon)\log n$

$$\lim_{n\to\infty} \mathbb{P}(X_\ell > 0) = 0 \tag{1.2}$$

It is known that this equality automatically holds when $\mathbb{E}(X_\ell) = o(1)$, and thus we want to prove this is true for given constraint on ℓ. Based on the statement (i), we have $\mathbb{E}(X_\ell) = \binom{n}{\ell} 2^{-\binom{\ell}{2}}$. Furthermore, we have $\binom{n}{\ell} \leq n^\ell$ and $2^{-\binom{\ell}{2}} = 2^{-\frac{\ell(\ell-1)}{2}}$, and thus

$$\begin{aligned}
\mathbb{E}(X_\ell) &\leq n^\ell 2^{-\frac{\ell(\ell-1)}{2}} \\
&\leq n^{\frac{2\ell}{2}} 2^{-\frac{\ell(\ell-1)}{2}} \\
&\leq \left(n^2 2^{-(\ell-1)}\right)^{\frac{\ell}{2}} \\
&\leq \left(2^{2\log n} 2^{-(\ell-1)}\right)^{\frac{\ell}{2}} \\
&\leq \left(\frac{2^{2\log n}}{2^{\ell-1}}\right)^{\frac{\ell}{2}}
\end{aligned} \tag{1.3}$$

For $\ell \geq (2+\varepsilon)\log n$, we have

$$\begin{aligned}
\frac{2^{2\log n}}{2^{\ell-1}} &\leq \frac{2^{2\log n}}{2^{(2+\varepsilon)\log n - 1}} \\
&\leq 2^{2\log n - (2+\varepsilon)\log n + 1} \\
&\leq 2^{1-\varepsilon\log n} \\
&\leq \frac{2}{n^\varepsilon}
\end{aligned}$$

Therefore,

$$\left(\frac{2^{2\log n}}{2^{\ell-1}}\right)^{\frac{\ell}{2}} \leq \left(\frac{2}{n^\epsilon}\right)^{\frac{\ell}{2}}$$

It follows from the inequality (1.3) that

$$\mathbb{E}(X_\ell) \leq \left(\frac{2}{n^\epsilon}\right)^{\frac{\ell}{2}}$$

and thus

$$\lim_{n\to\infty} \mathbb{E}(X_\ell) \leq \lim_{n\to\infty} \left(\frac{2}{n^\epsilon}\right)^{\frac{\ell}{2}} \tag{1.4}$$

When n goes to infinity and $\varepsilon > 0$, it is obvious that $0 < \frac{2}{n^\varepsilon} < 1$. Hence, when n goes to infinity, $\varepsilon > 0$, and $\ell \geq (2+\varepsilon)\log n$, we have

$$\begin{aligned}
\left(\frac{2}{n^\varepsilon}\right)^{\frac{\ell}{2}} &\leq \left(\frac{2}{n^\varepsilon}\right)^{\frac{(2+\varepsilon)\log n}{2}} \\
&\leq \left(2n^{-\varepsilon}\right)^{\frac{(2+\varepsilon)\log n}{2}} \\
&\leq 2^{\frac{(2+\varepsilon)\log n}{2}} n^{-\varepsilon\frac{(2+\varepsilon)\log n}{2}} \\
&\leq n^{\frac{2+\varepsilon}{2}} n^{-\varepsilon\frac{(2+\varepsilon)\log n}{2}} \\
&\leq n^{\frac{2+\varepsilon}{2}(1-\varepsilon\log n)} \\
&\leq n^{-\frac{2+\varepsilon}{2}(\varepsilon\log n - 1)}
\end{aligned}$$

It follows from the inequality (1.4) that

$$\lim_{n\to\infty} \mathbb{E}(X_\ell) \leq \lim_{n\to\infty} n^{-\frac{2+\varepsilon}{2}(\varepsilon \log n - 1)} = 0$$

This implies that $\mathbb{E}(X_\ell) = o(1)$, and thus the inequality (1.2) is satisfied, and the proof is finished.

□

Problem 2. In the settings of exercise 1, show the following statements:

(i)

$$\text{Var}(X_\ell) \leq \sum_{k=2}^{\ell} \binom{n}{\ell}\binom{\ell}{k}\binom{n-\ell}{\ell-k} 2^{-2\binom{\ell}{2}+\binom{k}{2}}$$

(ii) For every $\varepsilon > 0$:

$$\lim_{n\to\infty} \mathbb{P}(\omega \geq (2-\varepsilon)\log n) = 1$$

You may use without proof:

$$\lim_{n\to\infty} \sum_{k=2}^{\ell} \frac{\binom{\ell}{k}\binom{n-\ell}{\ell-k}2^{\binom{k}{2}}}{\binom{n}{\ell}} = 0$$

Proof.
It is known that

$$\text{Var}(X_\ell) = \mathbb{E}(X_\ell^2) - \mathbb{E}(X_\ell)^2$$

Recall from the Problem 1 that $X_\ell = \sum_{s\in S} X_\ell^{(s)}$ where S is the set of all subsets in vertices of G with cardinality ℓ. Therefore,

$$\begin{aligned}
\text{Var}(X_\ell) &= \mathbb{E}\left(\left(\sum_{s\in S} X_\ell^{(s)}\right)^2\right) - \mathbb{E}\left(\sum_{s\in S} X_\ell^{(s)}\right)^2 \\
&= \mathbb{E}\left(\sum_{s\in S} X_\ell^{(s)} \sum_{s'\in S} X_\ell^{(s')}\right) - \sum_{s\in S} X_\ell^{(s)} \sum_{s'\in S} X_\ell^{(s')} \\
&= \sum_{s,s'\in S} \left(\mathbb{E}(X_\ell^{(s)} X_\ell^{(s')}) - \mathbb{E}(X_\ell^{(s)})\mathbb{E}(X_\ell^{(s')})\right) \qquad (2.1)
\end{aligned}$$

Recall that $\mathbb{E}(X_\ell^{(s)}) = \mathbb{P}(A_\ell^{(s)})$ where $A_\ell^{(s)} = \{G \in \Omega_n |\ G_s \text{ is } K_\ell\}$, the equation (2.1) is equivalent to

$$\text{Var}(X_\ell) = \sum_{s,s'\in S} \left(\mathbb{P}(A_\ell^{(s)} \& A_\ell^{(s')}) - \mathbb{P}(A_\ell^{(s)})\mathbb{P}(A_\ell^{(s')})\right) \qquad (2.2)$$

If $s \cap s' = \emptyset$, then the events A_s and $A_{s'}$ are independent. Moreover, this also holds when $|s \cap s'| = 1$ since they do not share any common edges. Therefore, if $|s \cap s'| \leq 1$, then $\mathbb{P}(A_\ell^{(s)} \& A_\ell^{(s')}) = \mathbb{P}(A_\ell^{(s)})\mathbb{P}(A_\ell^{(s')})$, and thus $\mathbb{P}(A_\ell^{(s)} \& A_\ell^{(s')})$ and $\mathbb{P}(A_\ell^{(s)})\mathbb{P}(A_\ell^{(s')})$ cancel each other in the equation (2.2) when $|s \cap s'| \leq 1$. By this observation, the equation (2.2) is equivalent to

$$\text{Var}(X_\ell) \leq \sum_{s,s'\in S, |s\cap s'|\geq 2} \mathbb{P}(A_\ell^{(s)} \& A_\ell^{(s')}) \qquad (2.3)$$

For arbitrary two sets $s, s^{'} \in S$ such that $|s \cap s^{'}| = k$ where $2 \leq k \leq \ell$, and a graph $G \in \Omega_n$, the graphs G_s and $G_{s^{'}}$ are K_ℓ only when there are edges between any pair of distinct vertices in G_s, and $G_{s^{'}}$ respectively. That is, such $2\binom{\ell}{2}$ edges must be presented in G. It is worth to notice that s and $s^{'}$ shares k common vertices, and thus there are $\binom{k}{2}$ common edges between G_s and $G_{s^{'}}$ that are counted twice in $2\binom{\ell}{2}$. Therefore, we only need $2\binom{\ell}{2} - \binom{k}{2}$ edges in G that are incident to vertices in s and $s^{'}$ so that G_s and $G_{s^{'}}$ are K_ℓ. Recall that the probability to choose a graph $G \in \Omega_n$ having a specific edge is $1/2$. Therefore, the probability to choose a graph $G \in \Omega_n$ such that G_s and $G_{s^{'}}$ are K_ℓ when $|s \cap s^{'}| = k$ is $2^{-2\binom{\ell}{2}+\binom{k}{2}}$

Now, we want to calculate the number of ways to pick $s, s^{'} \in S$ such that $|s \cap s^{'}| = k$ for $2 \leq k \leq \ell$. Firstly, we want to arbitrarily pick s, which is a subset of vertices in G with cardinality ℓ, and thus there are $\binom{n}{\ell}$ ways to do so. For each set s, we can pick any k vertices in s that we want $s^{'}$ to share with, and there are $\binom{\ell}{k}$ ways to do this. For each selection of k vertices in s, we now want to construct $s^{'}$ by including these k vertices, as we want $|s \cap s^{'}| = k$, and pick $\ell - k$ vertices left from the vertices of G that are not in s, and thus there are $\binom{n-\ell}{\ell-k}$ ways to construct $s^{'}$. Therefore, there are $\binom{n}{\ell}\binom{\ell}{k}\binom{n-\ell}{\ell-k}$ ways to pick $s, s^{'} \in S$ such that $|s \cap s^{'}| = k$ for $2 \leq k \leq \ell$.

Based on the observations described above, by the equation (2.3), we have

$$\text{Var}(X_\ell) \leq \sum_{k=2}^{\ell} \binom{n}{\ell}\binom{\ell}{k}\binom{n-\ell}{\ell-k} 2^{-2\binom{\ell}{2}+\binom{k}{2}}$$

In order to prove the statement (ii), we want to show the following equality holds for $\ell \geq (2-\epsilon)\log n$

$$\lim_{n\to\infty} \mathbb{P}(X_\ell = 0) = 0 \tag{2.4}$$

It is known that

$$\mathbb{P}(X_\ell = 0) \leq \frac{\text{Var}(X_\ell)}{\mathbb{E}(X_\ell^2)} \tag{2.5}$$

By the statement (i) of the first and this problem, we have

$$\begin{aligned}
\frac{\text{Var}(X_\ell)}{\mathbb{E}(X_\ell)^2} &\leq \frac{\sum_{k=2}^{\ell}\binom{n}{\ell}\binom{\ell}{k}\binom{n-\ell}{\ell-k}2^{-2\binom{\ell}{2}+\binom{k}{2}}}{\binom{n}{\ell}^2 2^{-2\binom{\ell}{2}}} \\
&\leq \frac{\binom{n}{\ell}2^{-2\binom{\ell}{2}}\sum_{k=2}^{\ell}\binom{\ell}{k}\binom{n-\ell}{\ell-k}2^{\binom{k}{2}}}{\binom{n}{\ell}^2 2^{-2\binom{\ell}{2}}} \\
&\leq \sum_{k=2}^{\ell}\frac{\binom{\ell}{k}\binom{n-\ell}{2}2^{\binom{k}{2}}}{\binom{n}{\ell}}
\end{aligned} \tag{2.6}$$

By the two equalities: (2.5) and (2.6), we have

$$\lim_{n\to\infty} \mathbb{P}(X_\ell = 0) \leq \lim_{n\to\infty} \sum_{k=2}^{\ell}\frac{\binom{\ell}{k}\binom{n-\ell}{\ell-k}2^{\binom{k}{2}}}{\binom{n}{\ell}} = 0$$

Therefore, the equality (2.4) holds, and the proof is finished. □